Bond Total Return Swaps: Theory, Pricing & Practice

Burgess (2024) – SSRN 5024091 – Working Notes

Setting and Goal

A *Bond Total Return Swap* (Bond TRS) is a bilateral OTC contract in which the *payer* – who economically holds the underlying reference bond – passes the bond's full economic return (coupons plus mark-to-market performance) to the *receiver*, who in exchange pays a stream of funded interest, typically a floating index plus a fixed spread. If the bond defaults, the contract is unwound and the receiver pays the loss-given-default (LGD) to the payer.

The paper is presented as a slide deck in two parts:

- **Part One – Theory:** contract motivation, key features, and the constant-units vs. constant-notional conventions.
- **Part Two – Pricing & Practice:** the closed-form PV decomposition (bond coupons, performance, funding, LGD), bond-price projection from repo, the par funding spread, and an approximate pricing methodology.

The output of these working notes is a complete, library-ready formula stack for valuing a Bond TRS together with its breakeven funding spread, plus enough information to validate against the companion papers (Lou for the funding/discounting subtleties, the CSA paper for the choice of discount curve, Pucci for the bond / CMT interaction).

Cast of objects.

- $B(t)$ – dirty price of the reference bond at time t (per unit face value, in percent or decimals; we use decimals below).
- N_B – *bond* notional, $N_B = (\text{Units}) \times (\text{Face Value})$.
- N_C – *cash* (funding) notional. For an at-the-money trade with units fixed at t_0, $N_C = N_B \cdot B(t_0)$.
- r – bond coupon rate (fixed); τ_i – year fraction of coupon period $(t_{i-1}, t_i]$.
- F_j – floating index fixing for funding period j (e.g. 3M SOFR or compounded SOFR for the period); s – funding spread.
- $P(t_0, t_i)$ – riskfree (OIS) discount factor.
- $Q(t_0, t_i)$ – survival probability of the bond issuer to t_i (under $\mathbb{Q}$), bootstrapped from the issuer's CDS curve.
- $\widetilde{P}(t_0, t_i) \equiv Q(t_0, t_i)\, P(t_0, t_i)$ – *risky discount factor.*
- RR – recovery rate of the bond (typical assumption 40% for senior subordinated bonds in developed markets); LGD = 1 − RR.
- $\varphi \in \{+1, -1\}$ – sign convention: $\varphi = +1$ when receiving the bond coupons, $\varphi = -1$ when paying them.

1 Part One – Theory

1.1 Trade economics and rationale

The bond TRS replicates a leveraged purchase of the underlying bond without an outright cash outlay or balance-sheet treatment as a bond position. The payer holds (or synthetically replicates) the bond and forwards every income stream to the receiver:

- bond coupons paid through;
- mark-to-market performance: any rise in $B(t)$ is paid by the payer to the receiver, any fall is paid in the opposite direction;
- if the issuer defaults, the contract is unwound and the receiver pays LGD $= 1 -$ RR on the bond notional to the payer.

Why trade a TRS rather than buy the bond? Capital efficiency, leverage, and synthetic market access offset by counterparty risk, interest-rate sensitivity to both the funding leg and the bond performance, and the regulatory off-balance-sheet treatment.

1.2 Trade features

The contract is OTC and customisable. The slide-deck enumerates nine features that drive both pricing and risk:

1. OTC, customisable, highly bespoke.
2. Notional convention: *constant units* (fixed number of bonds) or *constant notional* (fixed cash amount).
3. Receive bond coupons and bond performance.
4. Pay fixed or floating funding.
5. Both interest-rate and credit risk (to the bond and to the counterparty / arranger).
6. Receiver exposed to LGD on bond default.
7. Bond performance projections are linked to repo rates.
8. Performance can be paid periodically or at maturity only.
9. Periodic performance payments reduce PFE and XVA costs relative to a bullet maturity payment.

1.3 Constant units vs. constant notional

Constant units. The number of bond units (Units) is fixed at trade inception. The *cash* notional moves with the bond price: each period a notional exchange is required to keep the funding leg consistent with the current value of the bond units. Bond fixings $B(t_i)$ are required at each reset.

$$\text{Units} = \frac{\text{Cash Notional}}{\text{Face Value} \times B(\text{fixing})}, \tag{1}$$

$$\text{Coupon} = \text{Units} \times \text{Face Value} \times r \times \tau. \tag{2}$$

Numerical sanity (deck example). Cash Notional = 10,000,000, Face Value = 100, Bond fixing = 123.86%:

$$\text{Units} = \frac{10{,}000{,}000}{100 \times 1.2386} = 80{,}736, \qquad \text{Coupon} = 80{,}736 \times 100 \times 3.85\% \times 0.5 = 155{,}416.80. \tag{3}$$

Constant notional. The cash notional is fixed; the number of bond units floats with each new fixing via the same Equation (1). No notional exchanges are required between coupon dates – the adjustment is absorbed into the changing units count. This is operationally simpler and is the more common contractual convention on dealer trades.

Remark 1 (Pricing implication)**.** *The two conventions give the same instantaneous PV at trade inception but evolve differently afterwards. Constant-units locks the unit count and lets the funding leg DV01 vary with bond price; constant-notional locks the funding leg DV01 and lets the unit count vary. For library validation it is essential that both conventions are supported and that the regression test exercises both.*

2 Part Two – Pricing & Practice

2.1 High-level decomposition

The Bond TRS PV decomposes into four parts:

$$\boxed{\mathrm{PV(TRS)} \;=\; \varphi\Big[\underbrace{\mathrm{PV(Bond\ Coupons)}}_{\text{(i)}} + \underbrace{\mathrm{PV(Performance)}}_{\text{(ii)}} - \underbrace{\mathrm{PV(Funding)}}_{\text{(iii)}} - \underbrace{\mathrm{PV(LGD)}}_{\text{(iv)}}\Big]} \tag{4}$$

where $\varphi = +1$ when receiving bond coupons / paying funding, and $\varphi = -1$ for the opposite leg.

2.2 (i) Bond coupons leg

For a fixed-rate bond paying r on $N_B = \text{Units} \times \text{Face Value}$ over n coupon periods:

$$\mathrm{PV(Bond\ Coupons)} \;=\; \sum_{i=1}^{n} N_B\, r\, \tau_i\, \underbrace{Q(t_0,t_i)}_{P(\text{Survive})}\, \underbrace{P(t_0,t_i)}_{\text{Disc}} \;=\; \sum_{i=1}^{n} N_B\, r\, \tau_i\, \widetilde{P}(t_0,t_i). \tag{5}$$

Each coupon is discounted under the survival-adjusted (*risky*) discount factor: the cash flow is only received if the bond has not defaulted by the coupon date.

2.3 (ii) Bond performance leg

The performance is the change in (dirty) bond price across each coupon period. Two payment conventions are supported:

$$\text{Performance paid each coupon date:}\quad \mathrm{PV(Perf)} = \sum_{i=1}^{n} N_B\,\big[B(t_{i-1}) - B(t_i)\big]\,\widetilde{P}(t_0,t_i), \tag{6}$$

$$\text{Performance paid at maturity } T\text{:}\quad \mathrm{PV(Perf)} = N_B\,\big[B(t_0) - B(T)\big]\,\widetilde{P}(t_0,T). \tag{7}$$

The slide-deck note that the performance leg is the largest contributor to PV(TRS) in absolute value, so projection of the future bond prices $B(t_i)$ is the most quantitatively sensitive ingredient.

Sign of the performance. Equation (6) is written from the payer's perspective on the cash-flow direction (bond holder pays away mark-to-market change). For a bond that appreciates, $B(t_i) > B(t_{i-1})$, this term is negative for the payer and positive for the receiver: with $\varphi = +1$ (receiver of bond returns) the receiver collects the appreciation, consistent with the economic intent.

2.4 Bond price projection from repo

Because the future bond price $B(t_i)$ enters the PV under the *risk-neutral* measure (after risky discounting), the no-arbitrage forward bond price is implied by the bond's repo rate, which acts as the bond's drift / cost-of-carry:

Proposition 1 (Forward bond price from repo)**.** *Let B_t be the dirty bond price at t and r_{repo} the repo rate applicable to the bond on the forward period of length $\tau = T - t$. The simple-compounded forward dirty price is*

$$B_T \;=\; B_t\,(1 + r_{\text{repo}}\,\tau). \tag{8}$$

If a coupon c is paid during (t, T), repo interest must be deducted on the coupon proceeds for the residual period τ_2 between coupon date and T:

$$\boxed{B_T \;=\; B_t\,(1 + r_{\text{repo}}\,\tau_1) - c\,(1 + r_{\text{repo}}\,\tau_2),} \tag{9}$$

where $\tau_1 = T - t$ and $\tau_2 = T - t_{\text{coupon}}$.

The slide-deck writes (9) in the factorised form $B_T = B_t(1 + r\,\tau_1 - rc\,\tau_2)$ when the notation is reused but the economic content is identical: the forward dirty price is the spot price grown at the repo rate, less the future-valued coupons paid in between.

Remark 2 (Why repo and not OIS)**.** *Repo is the cost at which a market maker can finance the bond position; it is the bond's true cost of carry. Using OIS would ignore the bond-specific repo spread and produce a forward price that is not arbitrage-free against the cash-and-carry trade. This is exactly the same observation made in the T-Lock paper (Pucci, SSRN 3386521) and the CSA paper (SSRN 1777005506) for the special case of bond collateral.*

2.5 (iii) Funding leg

The funding leg pays a floating coupon on the cash notional N_C: $F_j + s$ over period τ_j, with m floating periods. The arranger funds the position, and the funding cash flows are also contingent on no counterparty default (subject to CSA / unwind conventions; the deck treats them as risky on the bond credit, which is consistent with the receiver paying funding only as long as the trade is alive):

$$\text{PV(Funding)} \;=\; \sum_{j=1}^{m} N_C\,(F_j + s)\,\tau_j\,\widetilde{P}(t_0, t_j). \tag{10}$$

2.6 (iv) Loss given default

If the bond defaults during period $(t_{i-1}, t_i]$, the contract is unwound and the receiver pays $\text{LGD} \cdot N_B = (1 - \text{RR}) \cdot N_B$ to the payer. The probability of default in that period is the incremental survival decay:

$$P(\text{default in } (t_{i-1}, t_i]) \;=\; Q(t_0, t_{i-1}) - Q(t_0, t_i). \tag{11}$$

The corresponding PV term, paid at t_i (assumed for simplicity – in the slide-deck convention) and discounted on the riskfree curve (the cash flow is conditional on default, not survival):

$$\text{PV(LGD)} \;=\; \sum_{i=1}^{n} N_B\,(1 - \text{RR})\,\big[Q(t_0, t_{i-1}) - Q(t_0, t_i)\big]\,P(t_0, t_i). \tag{12}$$

Remark 3 (Why $P(,)$ and not $\widetilde{P}(,)$ for the LGD term)**.** *The LGD cash flow is paid* when *the bond defaults: it is conditioned on default, not on survival. Hence the appropriate discount factor is the riskfree $P(,)$, with the default probability $[Q(,t_{i-1}) - Q(,t_i)]$ supplying the analogue of the survival weight. This is consistent with the CDS default-leg convention.*

2.7 Master pricing formula

Assembling (5), (6), (10), (12) into (4) yields the closed-form Bond TRS PV (with periodic performance payments, n bond and n funding periods aligned for clarity; an unequal-frequency version replaces the funding sum with $j = 1, \ldots, m$):

$$\boxed{\begin{aligned}\mathrm{PV(TRS)} \;=\; \varphi \Bigg[& \sum_{i=1}^{n} N_B\, r\, \tau_i\, \widetilde{P}(t_0, t_i) + \sum_{i=1}^{n} N_B \left[B(t_{i-1}) - B(t_i)\right] \widetilde{P}(t_0, t_i) \\ & - \sum_{j=1}^{m} N_C\, (F_j + s)\, \tau_j\, \widetilde{P}(t_0, t_j) \\ & - \sum_{i=1}^{n} N_B\, (1 - \mathrm{RR}) \left[Q(t_0, t_{i-1}) - Q(t_0, t_i)\right] P(t_0, t_i) \Bigg].\end{aligned}} \tag{13}$$

2.8 Breakeven (par) funding spread

The *par* funding spread p is the value of the funding spread s that sets $\mathrm{PV(TRS)} = 0$ at trade inception (an at-the-money trade is free to enter, like a vanilla swap). Solve (13) for s:

Proposition 2 (Par spread)**.** *The breakeven funding spread is*

$$\boxed{p \;=\; \frac{\mathrm{PV(TRS}\ \textit{with}\ s = 0)}{\sum_{j=1}^{m} N_C\, \tau_j\, \widetilde{P}(t_0, t_j)},} \tag{14}$$

where the denominator

$$A^{\mathrm{risky}} \;\equiv\; \sum_{j=1}^{m} N_C\, \tau_j\, \widetilde{P}(t_0, t_j) \tag{15}$$

is the risky annuity *of the funding leg.*

Equation (14) is the direct analogue of the par CDS spread formula: the numerator is the PV of all non-spread legs (bond coupons + performance − floating index payments − LGD) under sign φ, and the denominator is the survival-weighted PV01 of the funding leg. The same structural identity makes both the CDS and the TRS *linear* in their respective spreads.

3 Approximate Pricing Methodology (Appendix of Deck)

The deck closes with a simplified valuation method that avoids explicit modelling of the LGD term and reuses standard bond-pricing infrastructure. The approximation relies on the observation that for an investment-grade name the LGD-weighted default term in (12) is small, and that the periodic performance $\sum[B(t_{i-1}) - B(t_i)]$ telescopes to a difference of bond prices when paid at maturity.

1. **Bond coupons.** Compute the PV of coupons that fall within the TRS effective and maturity dates – a standard risky-bond cash-flow strip valuation.
2. **Bond performance.** Compute the performance from two risky bond prices:
   - Current bond price: price the bond as standard.
   - Forward bond price at T_{TRS}: price the bond *with all coupons up to the TRS maturity removed* from its cash-flow set. The result is the dirty price of the residual bond, equivalent to the forward price implied by no-arbitrage on the bond cash flows.

The performance is then the difference $B(t_0) - B(T_{\text{TRS}})$ discounted on the risky curve to T_{TRS}.

3. **Funding leg.** Standard floating-leg pricing under the risky annuity A^{risky} of (15).
4. **LGD.** For investment-grade reference bonds, can be neglected in a first-cut PV: $Q(t_0, t_i) - Q(t_0, t_{i-1}) \approx 0$. For sub-IG, retain the full (12).

Net cash-flow picture (deck illustration). Netting bond legs and funding leg, the approximation is equivalent to:

- Full bond cash flows to bond maturity (received under $\varphi = +1$).
- Negative notional exchange equal to the current bond price $B(t_0)$ paid at TRS maturity (the "buy back" of the residual bond).
- Funding leg coupons to TRS maturity.

This decomposition makes plain the sense in which a Bond TRS is *economically equivalent* to a leveraged forward purchase of the bond (cf. Lou (2018), Section 2.3).

4 Cross-References Within the Working Notes Set

Lou (TRS, SSRN 3217420). The Burgess closed-form decomposition (13) is the *slide-deck textbook* version of the same product whose XVA-aware pricing is treated rigorously in Lou. The Burgess formulas implicitly assume:

- Riskfree discounting on $P(t_0, t_i)$, i.e. a fully-collateralised TRS (Lou's perfect-CSA case, equivalent to the OIS regime of the CSA paper).
- No funding-cost wedge between dealer-funding rate and CSA rate – i.e. Lou's $r_b = r_c = r$, $r_w = r$, hence $cva = dva = cfa = dfa = 0$.
- Bond carry impounded as a repo rate via Proposition 1 – exactly Lou's "security leg should absorb the financing cost of the hedge".

A natural extension is to apply Lou's recursive trinomial tree on top of (13) to recover counterparty-risk and funding adjustments.

CSA paper (Anonymous, SSRN 1777005506). The choice of $P(t_0, t_i)$ in (13) is a non-trivial one: which curve? The CSA paper is the canonical reference for the answer. Burgess uses "the" OIS curve implicitly; in the multi-CSA regime the discount factor is the curve dictated by the CSA collateral choice (cash CSA in funding currency $\Rightarrow$ OIS in that currency; bond collateral $\Rightarrow$ repo on the posted bond plus haircut). For the Bond TRS specifically, dealers usually post cash, so OIS is the standard choice and Equation (13) stands as written.

Pucci T-Lock (SSRN 3386521). The forward bond price derived in Proposition 1 is the same object that anchors the T-Lock as a forward contract on the underlying benchmark bond. The repo-derived projection (9) is mathematically identical to the T-Lock forward formula; only the wrapping is different (a performance leg here, a forward payoff there).

Pucci CMT (SSRN 3387961). The default-aware machinery underlying the bond yield ($\widehat{\widehat{D}}$, partial filtration modelling) is the natural framework in which to refine $Q(, t_i)$ when the issuer is correlated with the funding index. The Burgess deck assumes deterministic $Q(,)$ bootstrapped from CDS; if this assumption is violated, the Pucci CMT framework supplies the correction.

5 Validation Notes for the Pricing Library

The targeted validation exercises below isolate one feature of the Bond TRS at a time. Each is a self-contained pass that should match hand calculation or a reduced-form benchmark.

5.1 Pure-coupon strip

1. **No-default sanity.** Set $Q(t_0, t_i) \equiv 1$ for all i, RR irrelevant. Equation (5) should collapse to a riskfree fixed-coupon strip and match the library's bond-pricer for a bond with the same coupon rate and dates, run on the OIS curve only.
2. **Constant hazard rate.** Set $Q(t_0, t) = e^{-\lambda(t-t_0)}$ for some constant λ. The risky discount $\widetilde{P}(,) = e^{-\lambda\tau}P$ should be reproducible by feeding a flat CDS spread of $\lambda(1 - \mathrm{RR})$ to the library's CDS-implied hazard bootstrapper. Verify both PVs agree to numerical tolerance.

5.2 Performance leg

3. **Telescoping check.** For the periodic performance (6), in the limit of a flat risky discount curve ($\widetilde{P}(t_0, t_i) \equiv \overline{P}$ for all i), the sum collapses to $N_B[B(t_0) - B(t_n)] \cdot \overline{P}$, matching (7) on a single-period bullet trade.
4. **Forward bond price reproduction.** Generate $B(t_i)$ from (9) on the library's repo curve. Then independently reprice the bond at t_i using the library's bond pricer with the curve at t_0 rolled forward. The two forward prices should match to within the tenor spacing of the underlying repo curve.
5. **Constant-units vs. constant-notional.** At t_0, both conventions must produce identical PV. At $t_1 > t_0$ after a bond price move, run the library's mark-to-market under both conventions and check that the deltas match the analytic expectation: constant-units MTM = Units $\times \Delta B(t_1)$ plus interest accrual; constant-notional MTM = implied via reset of Units.

5.3 Funding leg and par spread

6. **Risky annuity.** On a flat survival curve and flat OIS curve, $A^{\text{risky}} = N_C \sum \tau_j e^{-\lambda t_j} e^{-r t_j}$. Library should match this to machine precision for a single-curve test.
7. **Par spread linearity.** The par spread p in (14) is by construction *linear* in the numerator. Bump the bond coupon r by Δr and verify that p moves by $\Delta r \cdot \frac{\sum N_B \tau_i \widetilde{P}(t_0, t_i)}{A^{\text{risky}}}$.
8. **Par spread sign.** For a bond appreciating over the TRS life ($B(T) > B(t_0)$, with bond coupons modest), the receiver of bond returns should be willing to pay a positive funding spread: $p > 0$. Conversely a depreciating bond gives $p < 0$ (negative spread, i.e. the dealer pays the receiver).

5.4 LGD term

9. **Recovery sensitivity.** $\partial\mathrm{PV(LGD)}/\partial\mathrm{RR}$ should be a positive number equal to $-\sum_i N_B[Q(, t_{i-1}) - Q(, t_i)]P(, t_i)$ (it reduces the magnitude of the negative term in (13)). Bump RR from 40% to 50% and check the bump matches the closed-form derivative.
10. **Cross-check against CDS.** A long-protection CDS on the same reference name pays $(1 - \mathrm{RR})$ on default; the analogous default-leg PV is

$$\mathrm{PV}^{\mathrm{CDS-def}} = (1 - \mathrm{RR}) \sum_i [Q(, t_{i-1}) - Q(, t_i)]P(, t_i),$$

which is the per-unit-notional version of (12). The library's CDS pricer should produce the identical number for the same hazard / discount inputs.

5.5 Numerical and limiting cases

11. **Zero-spread regression.** Set $s = p$ via (14); the resulting PV should be *identically* zero (to numerical tolerance). Useful as an end-to-end sanity check after any change to the curves or the discount logic.

12. **Approximate vs. exact.** For an investment-grade name with $\lambda T \ll 1$, the approximate methodology of Section 5 should match the exact (13) to within a known LGD-dropped error term. Quantify the drift between the two for a representative IG name (e.g. flat 50bp CDS, 5Y maturity) – this provides a regression fixture for future curve-engine changes.

13. **Stress: high-spread name.** Repeat the IG-vs-exact comparison for a sub-IG name (flat 500bp CDS). The approximation should now noticeably under- or over-estimate the PV depending on φ, and the drift should match $\varphi \cdot \mathrm{PV(LGD)}$ to within discretisation error.

6 Notation Cheat-Sheet

Symbol	Meaning
φ	$+1$ if receiver of bond return / payer of funding; -1 otherwise
N_B	Bond notional, = Units $\times$ Face Value
N_C	Cash (funding) notional
$B(t)$	Dirty bond price at t (per unit face value)
r	Bond fixed coupon rate
τ_i, τ_j	Year fractions for bond / funding period
F_j	Floating-rate fixing for funding period j (e.g. SOFR)
s, p	Funding spread (running) and par spread
$P(t_0, t)$	Riskfree (OIS) discount factor
$Q(t_0, t)$	Survival probability of the bond issuer
$\widetilde{P}(t_0, t)$	Risky discount factor, $Q(,t)P(t_0, t)$
RR	Recovery rate; LGD $= 1 -$ RR
r_{repo}	Repo rate of the bond (per period; bond-specific)
A^{risky}	Risky annuity of the funding leg

7 Bibliographic Cross-References

- Burgess, N. (2024). *Bond Total Return Swaps: Theory, Pricing & Practice.* SSRN 5024091. (The slide-deck summarised here.)
- Burgess, N. (n.d.). *Low Latency Interest Rate Markets.* `github.com/nburgessx/SwapsBook` (companion book referenced by the deck).
- Lou, W. (2018). *Pricing Total Return Swap.* SSRN 3217420. (Companion paper in our set; pushes the funding-cost treatment beyond what the Burgess deck covers.)
- Anonymous (n.d.). *Why We Don't Discount at the "Risk-Free Rate": What Textbooks Get Wrong About Derivatives Pricing, and What Trading Desks Actually Do.* (Companion paper; governs the choice of curve for $P(,)$.)
- Pucci, A. (2019). *Hedging the Treasury Lock.* SSRN 3386521. (Companion paper; uses the same repo-implied forward bond price as Proposition 1.)

- Pucci, A. (2014). *Constant Maturity Treasury Convexity Correction.* SSRN 3387961. (Companion paper; default-aware bond-yield framework relevant if the Burgess hazard assumption $Q(,)$ deterministic is relaxed.)